

Headless mode

Canon R7 EXR Converter - FVFX v2.3.0 · command-line reference for batch jobs, render-farm scripts and pipeline tools

Headless mode runs the same converter as the app, with no window. It starts whenever `--convert` is on the command line. It uses the app's default settings, so the output is identical to pressing Convert with Advanced left at its defaults.

Synopsis

```
"Canon R7 EXR Converter v2.3.0.exe" --convert CLIP [CLIP ...] --out DIR [--log FILE] [--prores] [--no-exr]
```

Options

Option	Meaning
<code>--convert CLIP ...</code>	One or more clip files. Every argument after it that does not start with <code>--</code> counts as a clip. Folders and wildcards are not expanded: list files (see the batch example).
<code>--out DIR</code>	Required. Output folder; created if missing.
<code>--log FILE</code>	Write progress and errors to FILE (overwritten each run). Essential with the exe: it is a windowed program and has no console. The folder must already exist.
<code>--prores</code>	Also write a ProRes .mov (422 HQ, 10-bit).
<code>--no-exr</code>	Skip the EXR sequence. Use it with <code>--prores</code> to write ProRes only.

Fixed settings

Headless mode has no flags for the Advanced-tab options. It always uses:

Setting	Value
Camera colour space	Detected from the clip (needs exiftool; otherwise Canon Cinema Gamut)
Output colour space	ACEScg (AP1), scene linear
Size	Full (source resolution)
EXR	16-bit half, ZIP compression, frame numbers start at 0001
ProRes	422 HQ, 10-bit, with the clip's timecode and audio

For any other combination, use the app.

Output

```
DIR\<clip>\<clip>.0001.exr ...    EXR sequence (one folder per clip)
DIR\<clip>.mov                  ProRes, next to the EXR folder
```

Existing files with the same names are **overwritten without asking**. `<clip>` is the clip's file name without its extension.

ProRes clips highlights. ProRes stores integers, so scene-linear values above 1.0 (about 2.5 stops over grey) are cut off. On the R7 test clip used in development that was 16% of pixel values. Use the .mov for review and editorial, and composite from the EXRs.

Running the exe

The exe is a windowed program, so a shell that launches it does not wait for it to finish and gets no output. Make the shell wait, and read the result from the log and the exit code. The installer puts the exe in `%LOCALAPPDATA%\Programs\Canon R7 EXR Converter\`; the zip build runs from wherever it was unpacked.

Command Prompt / .bat

```
start "" /wait "Canon R7 EXR Converter v2.3.0.exe" ^
    --convert "D:\Shoot\A001.MP4" --out "D:\Shoot\EXR" --log "D:\Shoot\convert.log"
echo exit code %ERRORLEVEL%
```

PowerShell

```
$exe = "$env:LOCALAPPDATA\Programs\Canon R7 EXR Converter\Canon R7 EXR Converter v2.3.0.exe"
$p = Start-Process $exe -Wait -PassThru -ArgumentList `
    '--convert "D:\Shoot\A001.MP4" --out "D:\Shoot\EXR" --log "D:\Shoot\convert.log" --prores'
$p.ExitCode
```

From source (development checkout). Output goes to the console, so --log is optional:

```
.venv\Scripts\python.exe -m r7convert --convert A001.MP4 A002.MP4 --out EXR
```

Batch a folder, one clip per run

A run stops at the first clip that fails, and all clips are read before any are converted. Running one clip per invocation keeps one bad file from stopping the rest, and gives each clip its own log:

```
$exe = "$env:LOCALAPPDATA\Programs\Canon R7 EXR Converter\Canon R7 EXR Converter v2.3.0.exe"
$out = "D:\Shoot\EXR"
$log = New-Item -ItemType Directory -Force "D:\Shoot\logs"
Get-ChildItem "D:\Shoot\*.MP4" | ForEach-Object {
    $log = Join-Path $logs "$($_.BaseName).log"
    $a = "--convert `"$($_.FullName)`" --out `"$out`" --log `"$log`" --prores"
    $p = Start-Process $exe -ArgumentList $a -Wait -PassThru
    if ($p.ExitCode -ne 0) { Write-Warning "$($_.Name) failed - see $log" }
}
```

Log and exit codes

Exit code 0 means every clip converted; 1 means something failed, and the last log line starts with error:. A successful log:

```
Canon R7 EXR Converter v2.3.0
ffmpeg: C:\...\tools\ffmpeg\ffmpeg.exe
exiftool: C:\...\tools\exiftool\exiftool.exe
904A4288: 3840x2160, 135 frames
    135 frames written
```

Look out for exiftool: None (the gamut can't be detected, so Cinema Gamut is used), Colour space not found in clip and Warning: clip is ..., not Canon Log 3. None of these stop the run, so a script should check for them.

Gotchas

Situation	What happens
--out missing, or the --log folder missing	Unhandled error before logging starts: nothing reaches the log, and the exe may show an error dialog. Always pass --out, and create the log folder first.
Stopping a run	There is no cancel flag. Ending the process leaves partial output: the frames written so far and an unplayable .mov.
Trimmed test clips	Clips cut with ffmpeg -c copy lose Canon's metadata, so the gamut falls back to Cinema Gamut. Test with camera originals.
Speed	About 0.35 s per 4K frame (about 3 fps); a 5 s clip takes about 45 s. Running several instances at once rarely helps: the bottleneck is moving frames between processes, not the CPU.